

Whole-Number Division: Interpreting Remainders

Answer Key

Grades 4–5

Quick Answers

1. 8	4. 34	7. 6, 37	10. 53, 2
2. 2	5. 59	8. 138	
3. 15	6. 2	9. 307, 7	

Common wrong answers

10. *If they got 52: dropped the remainder instead of rounding up, leaving 2 marbles with no box*

1. Apples in bags of 6. Divide: $53 \div 6$. Since $6 \times 8 = 48$ and $53 - 48 = 5$, the quotient is 8 with remainder 5.

The stand sells only *full* bags, and 5 apples are not enough for another full bag — the remainder is dropped.

8 full bags

2. Stickers shared among 5 friends. Divide: $47 \div 5$. Since $5 \times 9 = 45$ and $47 - 45 = 2$, each friend gets 9 whole stickers and the remainder is 2.

The question asks what is left over, so the remainder itself is the answer.

2 stickers left over

3. Vans for 128 students. Divide: $128 \div 9$. Since $9 \times 14 = 126$ and $128 - 126 = 2$, the quotient is 14 with remainder 2.

Fourteen vans seat only 126 students; the 2 remaining students still need a seat, so one more van is needed — round the quotient up.

15 vans

4. Ribbon cut into 8-centimetre pieces. Divide: $275 \div 8$. Since $8 \times 34 = 272$ and $275 - 272 = 3$, the quotient is 34 with remainder 3.

The leftover 3 centimetres is shorter than a whole piece, so it cannot count as a piece — drop the remainder.

34 whole pieces

5. Eggs in cartons of 6. Divide: $350 \div 6$. Since $6 \times 58 = 348$ and $350 - 348 = 2$, the quotient is 58 with remainder 2.

Every egg must be in a carton, and 58 cartons hold only 348 eggs, so the last 2 eggs need one more carton — round up.

59 cartons

6. Trading cards, 9 to a page. Divide: $623 \div 9$. Since $9 \times 69 = 621$ and $623 - 621 = 2$, the quotient is 69 with remainder 2.

Rosa fills 69 full pages, and the 2 remaining cards go on the next page — the last page she uses is partly filled, and the remainder tells how many cards are on it.

2 cards on the last page

7. Buses for 237 students, 40 seats each. (a) Estimate: 237 rounds to 240 to the nearest ten, and $240 \div 40 = 6$, so about 6 buses. about 6 buses

(b) Exactly: $237 \div 40$ gives $40 \times 5 = 200$ with $237 - 200 = 37$ left, so 5 buses fill completely and the remainder rides on a sixth bus. The estimate of 6 buses was right, and the last bus is only partly full. 37 students on the last bus

8. Pencils in boxes of 9. Divide: $1248 \div 9$. Since $9 \times 138 = 1242$ and $1248 - 1242 = 6$, the quotient is 138 with remainder 6.

Only completely filled boxes are counted, and the 6 leftover pencils do not fill a box — drop the remainder. 138 full boxes

9. Books on shelves of 8. Divide: $2455 \div 8$. Since $8 \times 306 = 2448$ and $2455 - 2448 = 7$, the quotient is 306 with remainder 7.

(a) Every book must be on a shelf. 306 shelves hold 2448 books, so the 7 remaining books need one more shelf — round up. 307 shelves

(b) The last shelf holds only those leftover books, which is the remainder.

7 books on the last shelf

10. Find and fix the mistake — marbles in boxes of 6. Maya's division is correct: $6 \times 52 = 312$ and $314 - 312 = 2$, so $314 \div 6 = 52 \text{ R } 2$. Her *interpretation* is what went wrong: she reported the quotient, but 52 boxes hold only 312 marbles, which leaves 2 marbles with no box.

(a) The question asks how many boxes are needed so that every marble is boxed, so the remainder forces one more box — round up. 53 boxes

(b) That last box is not full; it holds only the leftover marbles, the remainder.

2 leftover marbles